

Project Requirement Document (PRD) Jatre-Namma – Digital Guide to the Village Fair

Android App Development using Modern Technologies

1. Project Overview

Jatre-Namma is a mobile application designed to digitally organize and guide visitors during village fairs ("Jatre"). These fairs are important cultural and social gatherings in rural Karnataka and attract large crowds from nearby villages and cities.

The application provides real-time schedules, safety information, navigation assistance, and cultural storytelling to improve visitor experience and public safety.

2. Problem Statement

Village Jatres are major cultural celebrations that include religious processions, wrestling competitions, dramas, cattle fairs, food stalls, and entertainment events.

However, visitors often face several challenges:

- Lack of access to event schedules
- Difficulty locating parking areas and first-aid centers
- Chaos in crowded environments
- No centralized lost & found support system
- Limited awareness about the cultural significance of the Jatre

These issues reduce convenience, safety, and accessibility for visitors.

3. Vision Statement

Jatre-Namma aims to become a "Digital Guide to the Village Fair" by blending traditional celebrations with modern technology.

The application captures the vibrant energy of the Jatre while ensuring:

- Organized event management
- Real-time visitor assistance
- Community engagement
- Public safety and emergency coordination
- Cultural preservation for younger generations

4. Core Features

4.1 Live Schedule

- Real-time updates for ongoing and upcoming events
- Examples: "Ratha starts at 4 PM", "Wrestling at 6 PM"
- Highlight the currently active event

4.2 Lost & Found

- Users can post details or photos of lost people/items
- Community-driven digital help desk
- Ability to mark cases as resolved

4.3 Parking & Safety

- Display parking zones and emergency routes
- Locate first-aid centers and police help desks

4.4 Cultural Stories

- Information about the legend and origin of the Jatre
- Educational and cultural storytelling content

5. User Flow

1. User opens the application
2. Home screen displays live Jatre activities
3. User checks ongoing event schedules
4. User accesses maps for parking or emergency support
5. User reports or searches for lost items/persons
6. User reads cultural stories and historical significance

6. Technical Requirements

Frontend:

- Android application using Jetpack Compose
- Vibrant festive design with traditional motifs

Backend:

- Firebase Realtime Database or Firestore for live updates
- Cloud storage for lost & found images

Maps & Navigation:

- Custom fairground map with temporary “Jatre Markers”
- Parking and safety zone indicators

Performance:

- Optimized loading for high traffic situations
- Fast UI response even during peak usage

7. UI/UX Requirements

- Festive and colorful interface design
- Traditional Karnataka-inspired visual elements
- Simple navigation for users of all age groups
- Quick loading screens and responsive layouts

8. Impact Goals

Community Resilience

Using technology to organize large traditional gatherings efficiently.

Cultural Continuity

Helping younger generations stay connected to village traditions and festivals.

Public Safety

Reducing confusion and improving emergency response during crowded events.

9. Success Criteria

- The Live Schedule must clearly highlight the current ongoing event
- The Lost & Found feature must support “Mark as Resolved” functionality
- Maps should accurately display temporary fair markers
- UI should handle simulated high traffic efficiently
- The application must load quickly and remain responsive

10. Suggested Future Enhancements

- Push notifications for event reminders
- Multilingual support (Kannada, English, Hindi)
- AI crowd monitoring and analytics
- Offline support for low-network areas
- Integration with local administration and emergency services

Prepared for academic project review and Android application planning.